

Coastal Zones and Estuaries

The Coastal Zone - Coastal Ocean extends from the high-tide line to the shelf break

- This region is responsible for nearly 30% of the total net oceanic primary production and nearly 90% of the global fish catch

Estuaries -Two major components involved:

- Transition from fresh (river) water to saline (ocean) water
- Tidal influence
- One definition:
 - “An estuary is a semi-enclosed coastal water body that extends to the effective limit of tidal influence, within which sea water is significantly diluted with freshwater from land drainage”

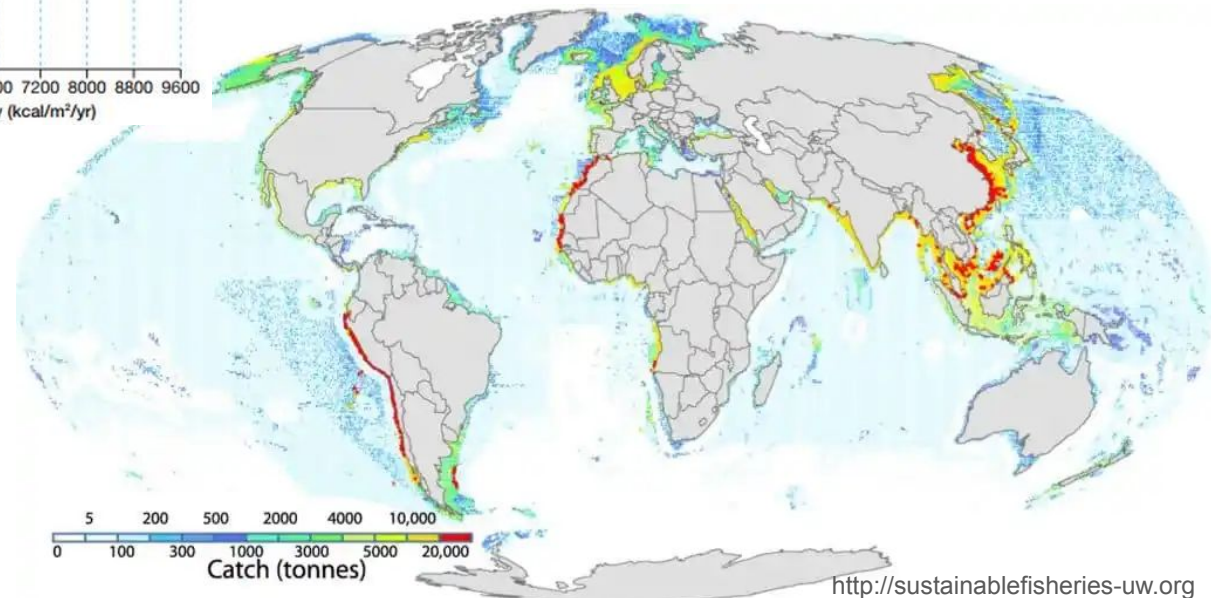
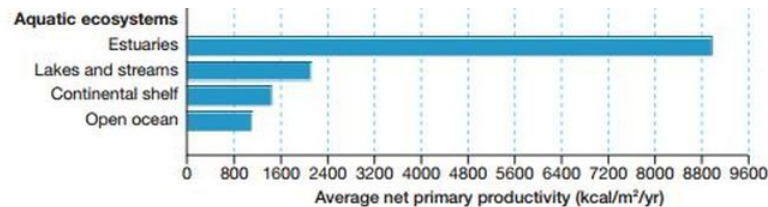
Coastal biogeochemistry

Learning objectives:

- A. Apply the concepts of organic matter production, remineralization and early diagenesis to eutrophication and nutrient cycling in coastal areas
- B. Understand how estuaries act as buffer zones between land and the open ocean
- C. Relate the biogeochemistry of coastal areas to their physical properties
- D. Learn about the stresses imposed to coastal areas by human activity

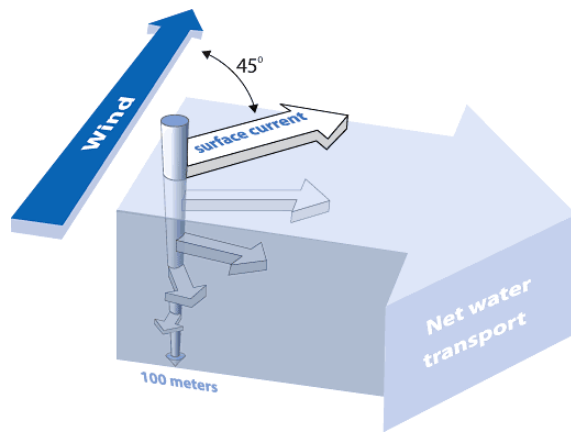
1. Productive areas

Coastal environments are responsible for nearly 30% of the total net oceanic primary production and nearly 90% of the global fish catch



2. Coastal upwelling

- Ekman transport:



- Friction + Coriolis means that net water transport is 90 deg. to the left or right of winds

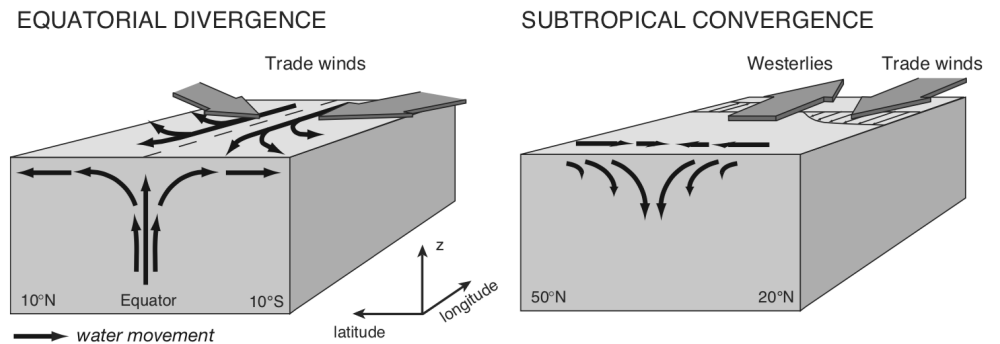


FIGURE 2.2.8: (a) Equatorial divergence and upwelling resulting from Ekman transport driven by trade winds [Thurman, 1990]. (b) Ekman downwelling resulting from convergent Ekman transport driven by the westerlies and trade winds.

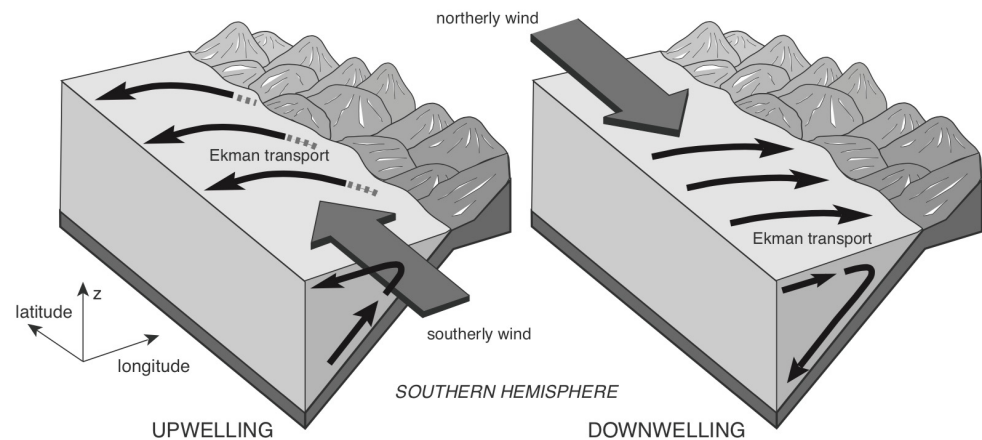
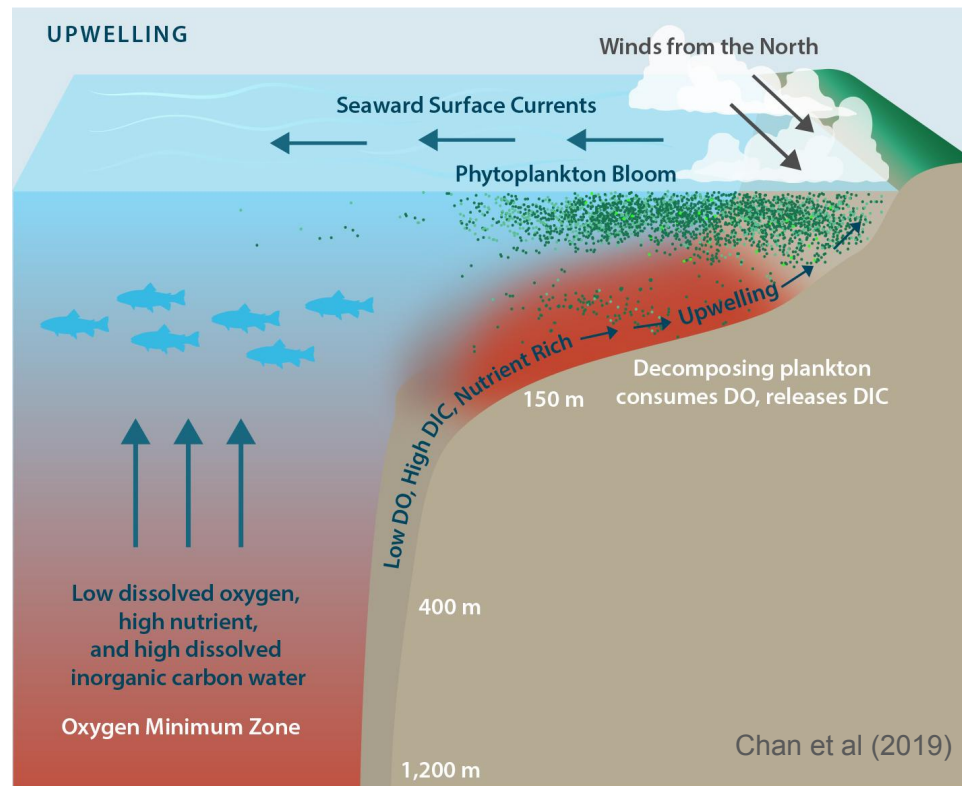


FIGURE 2.2.6: Ekman transport and associated upwelling and downwelling resulting from wind blowing parallel to shore [Thurman, 1990]

2. Upwelling along coasts

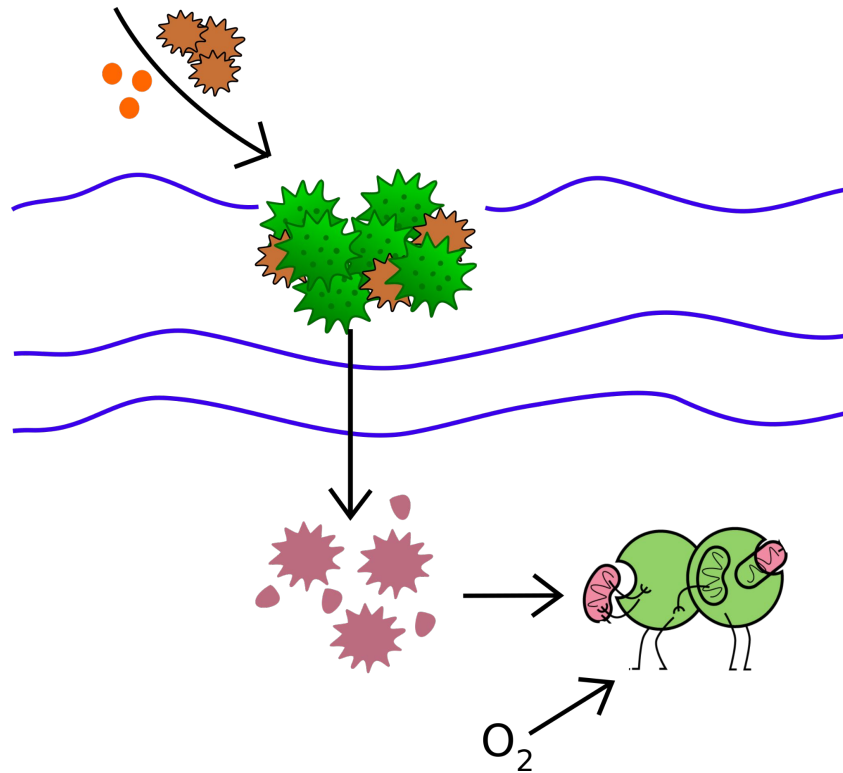
In coastal areas, nutrient cycling is *accelerated*

- Upwelling of low oxygen and nutrient rich waters
- High productivity and hence fisheries
- Further oxygen depletion through eutrophication

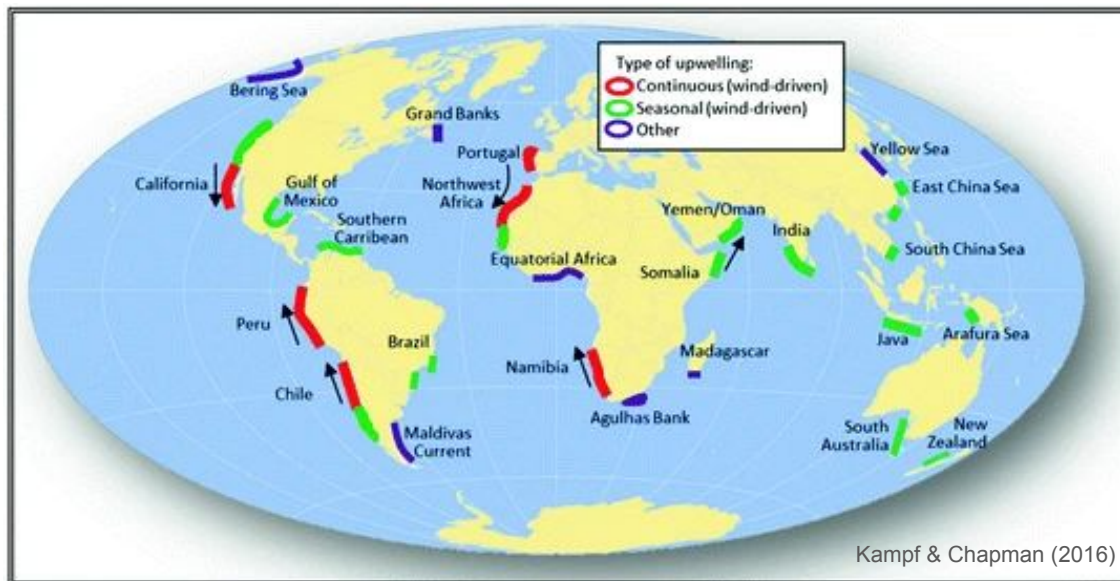


Eutrophication

1. Nutrient and organic matter input from land
2. Organic matter production
3. Settling of organic matter towards depth
4. Decomposition by aerobic bacteria (oxygen sink)

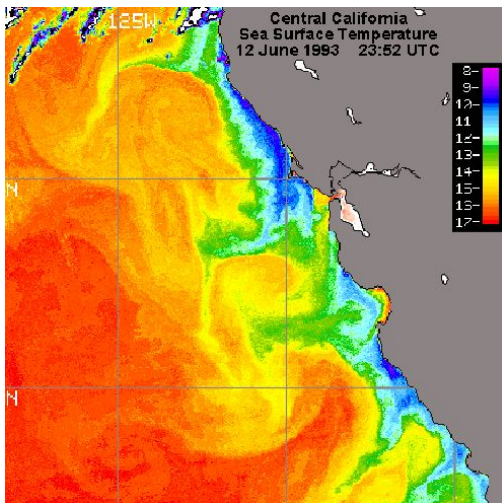


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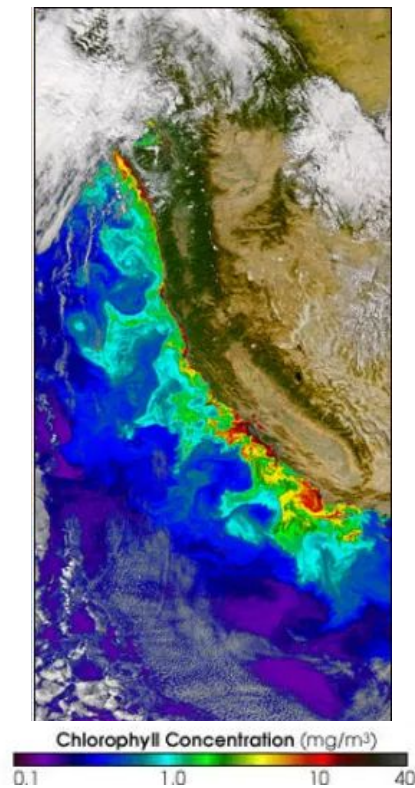


2. Upwelling along coasts

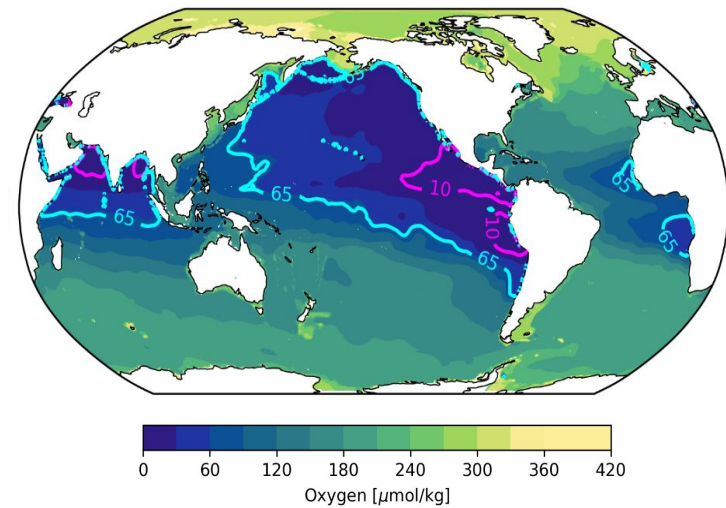
Surface temperature



Productivity

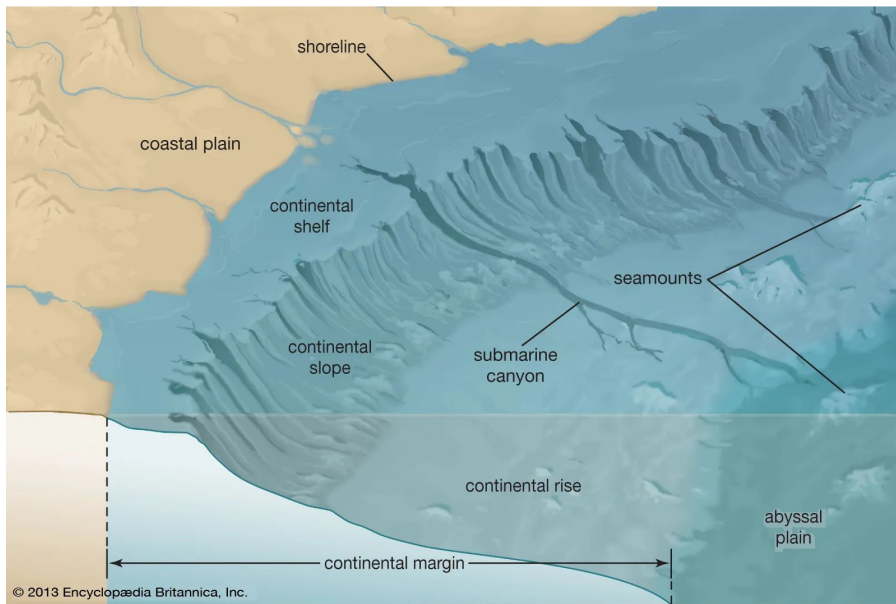


Oxygen minimum zones



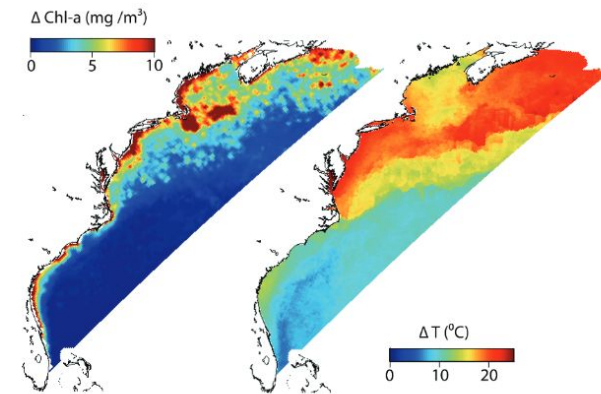
Oxygen concentration at the minimum oxygen level.
Data from the World Ocean Circulation Experiment.
Cyan line: hypoxia. Pink line: anoxia.

3. Continental shelves



<https://www.britannica.com/science/continental-shelf>

Charaterized by large seasonal variations



Fennel (2010)

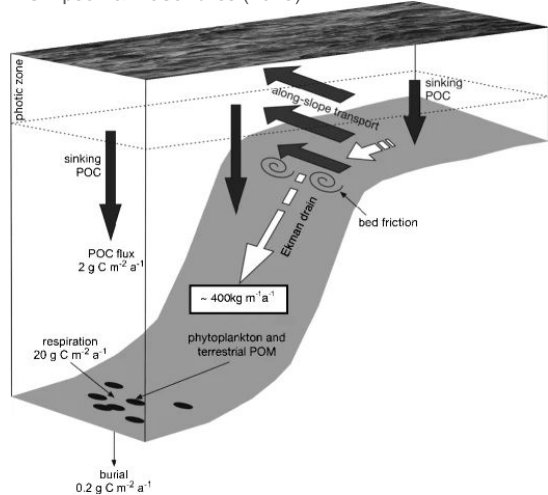
3. Continental shelves

- Nutrient sources:

- From land (river export)
- Upwelling
- Remineralization

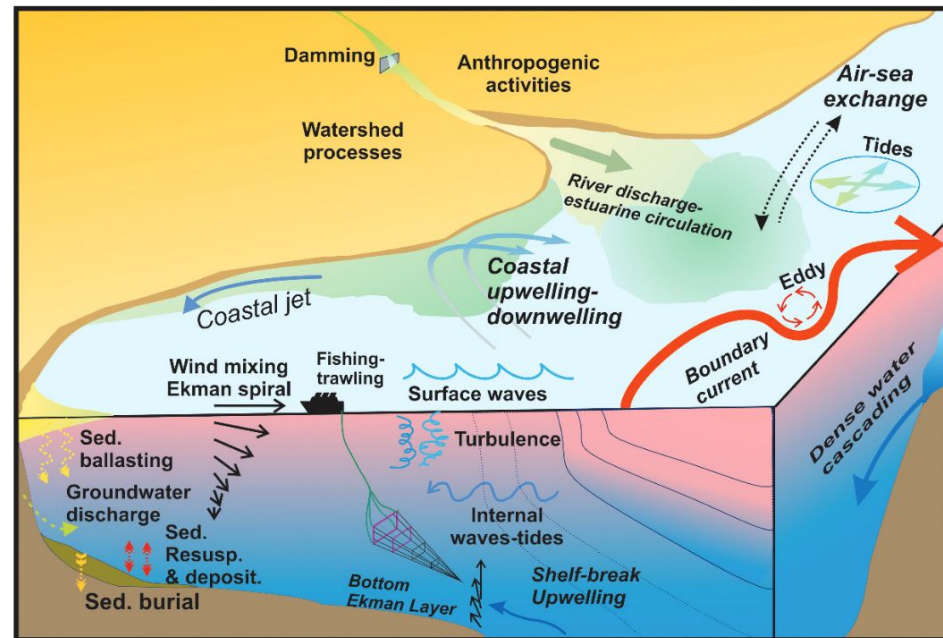
High productivity

Simpson & McCandless (2013)



- Cross-shelf exchanges:

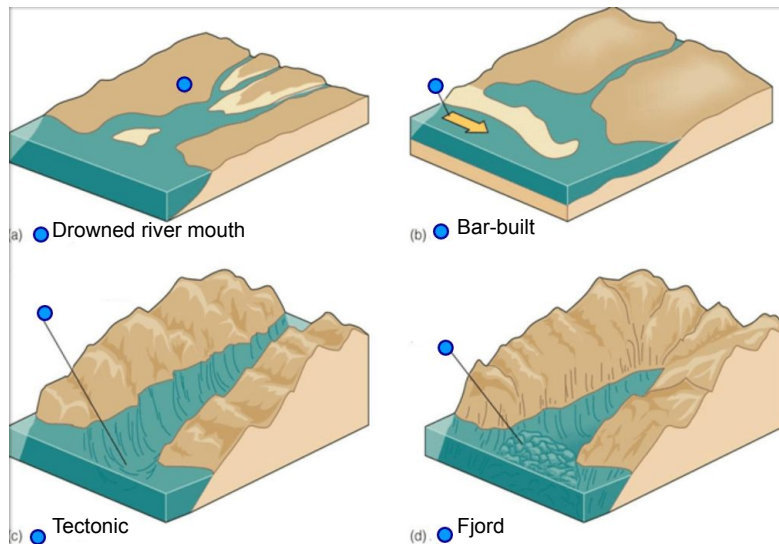
- Interaction with boundary currents
- Upwelling ($>$ river input)
- Eddy shedding and filaments
- Ekman drain



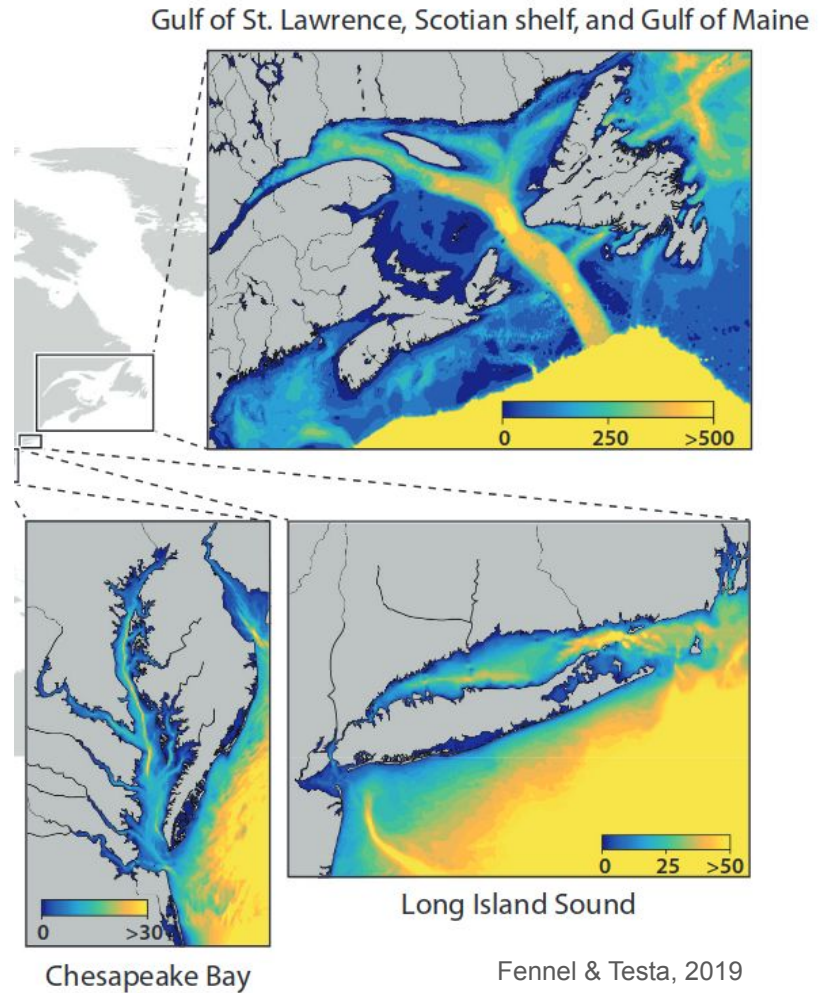
Liu et al. (2010)

4. Estuaries

Where the river meets the sea



Graphonic 2016



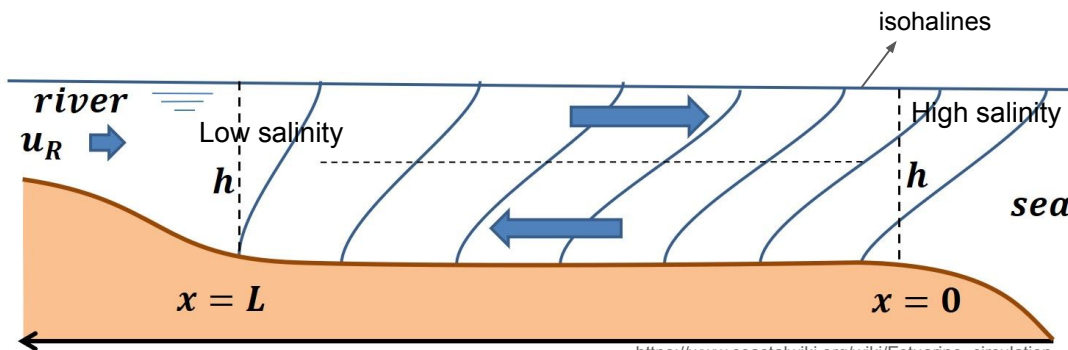
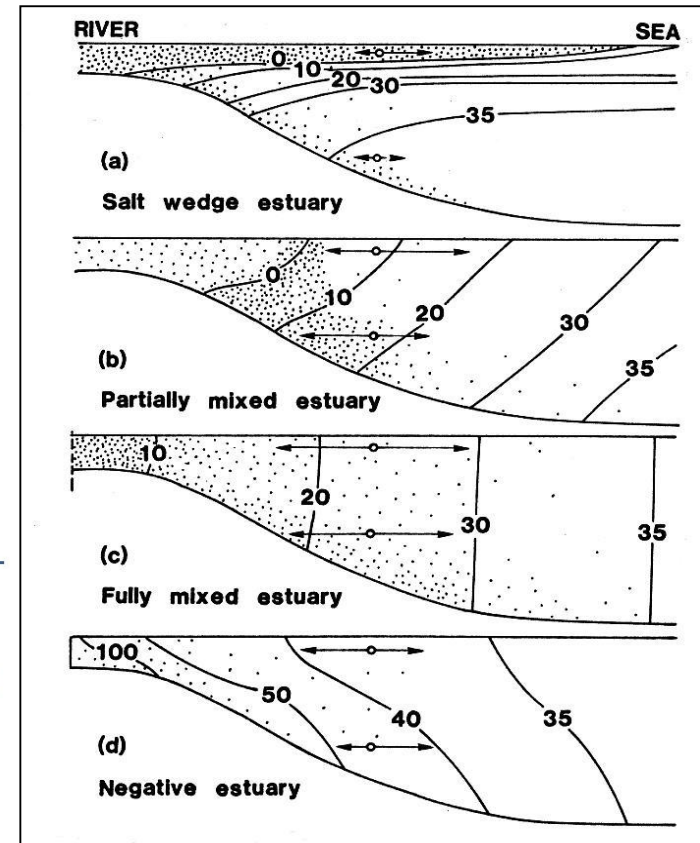
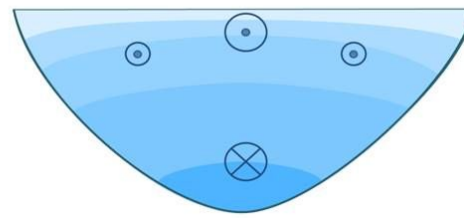
4. Estuaries

Gravitational (or estuarine) circulation

Denser waters sink,
intruding inland

Seaward flow at the
surface to balance out

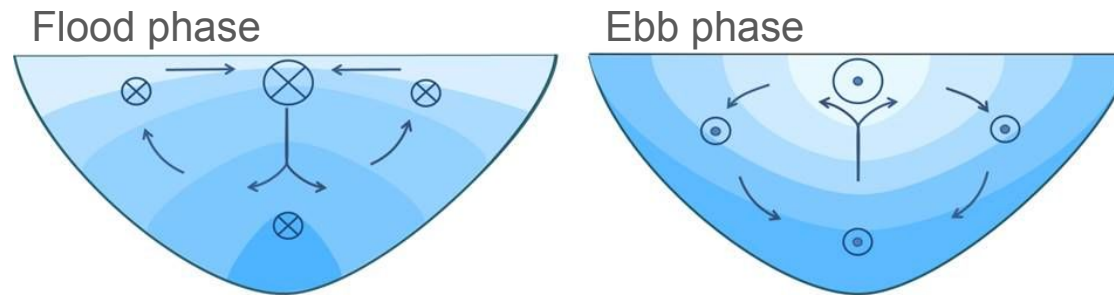
Tides



https://www.coastalwiki.org/wiki/Estuarine_circulation

4. Estuaries

Gravitational (or estuarine) circulation: tides



1. Tides + estuarine circulation = shear
2. Shear induces turbulent mixing
3. Mixing maintains the gravitational circulation

4. Estuaries

Gravitational (or estuarine) circulation: jets

Coriolis effect:

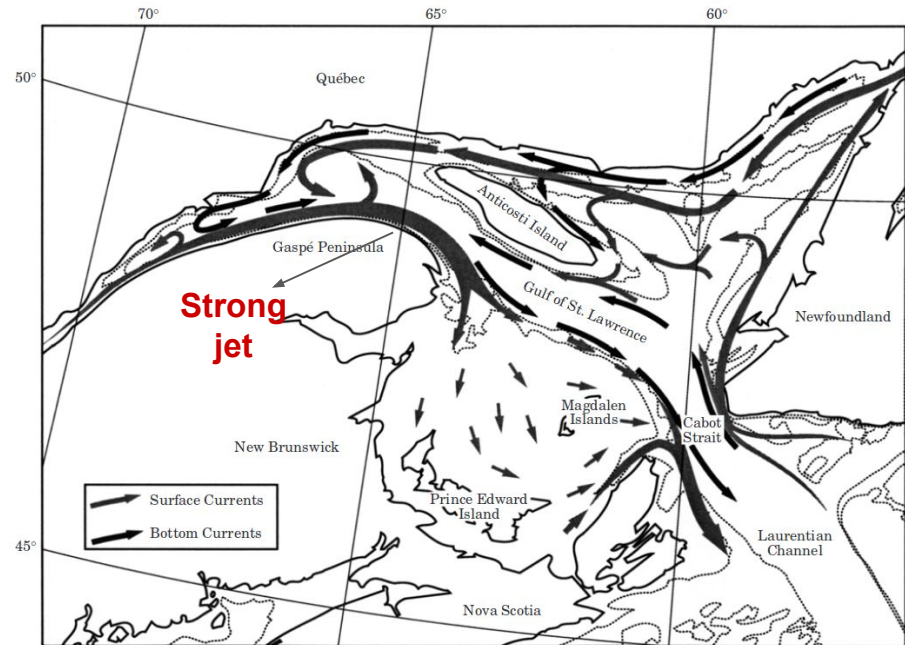
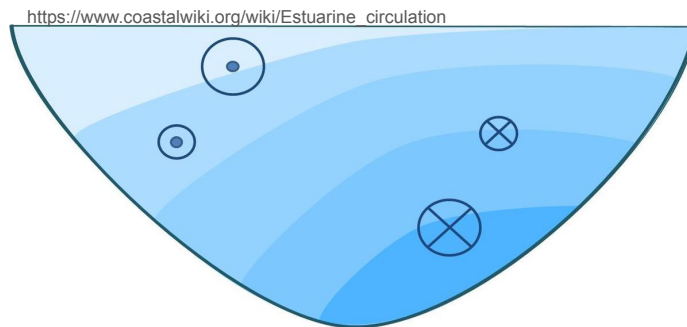


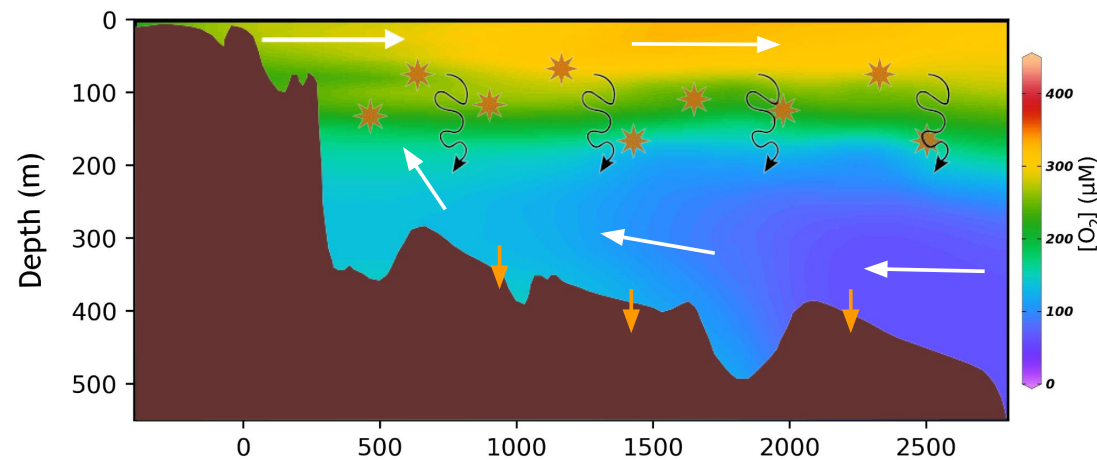
FIGURE 2. Mean summer near-surface and near-bottom currents in the Gulf of St. Lawrence, based on the model solution of Han *et al.* (1999).
Savenkoff *et al.* (2001)

4.2 Nutrient cycling in estuaries

- Large river nutrient and OM export
- Nutrient upwelling
- **High production** in the surface
- Some export downstream
- Settling of OM and decomposition
- Burial in sediments

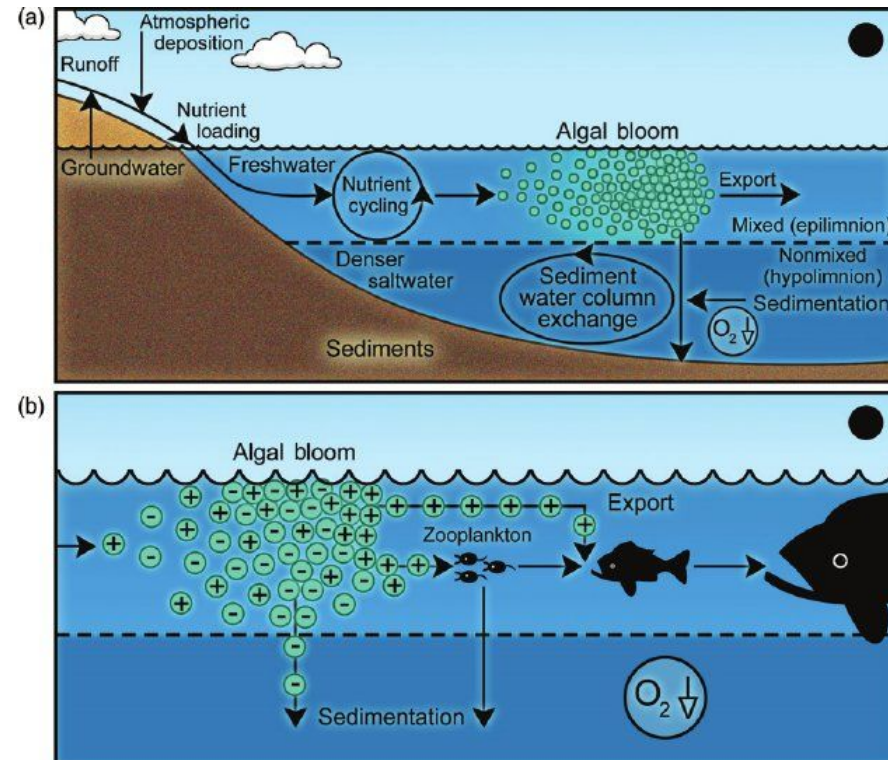
Estuaries act as **nutrient traps**

(Hence remove excess nutrients and do carbon sequestration before the waters reach the sea)



4.2 Nutrient cycling in estuaries

OM decomposition can lead to hypoxia (eutrophication)

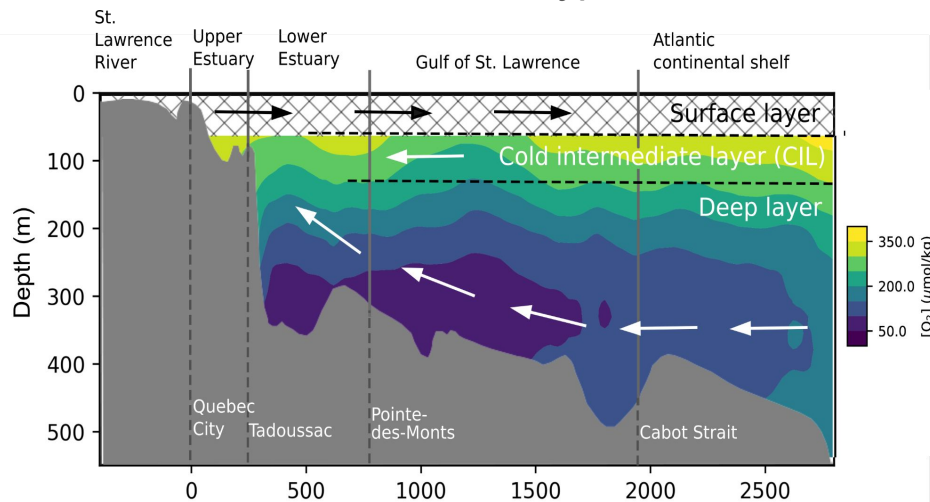


Pearl (2011), Phytoplankton Ecology and Trophic Dynamics in Coastal Waters

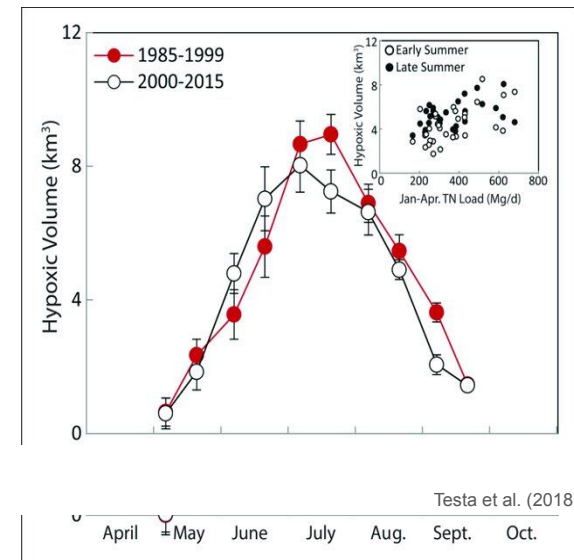
4.2 Nutrient cycling in estuaries

Hypoxia duration

Permanent stratification/hypoxia



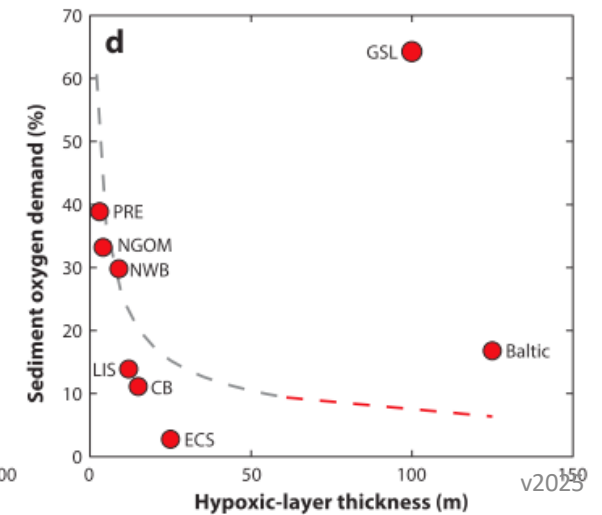
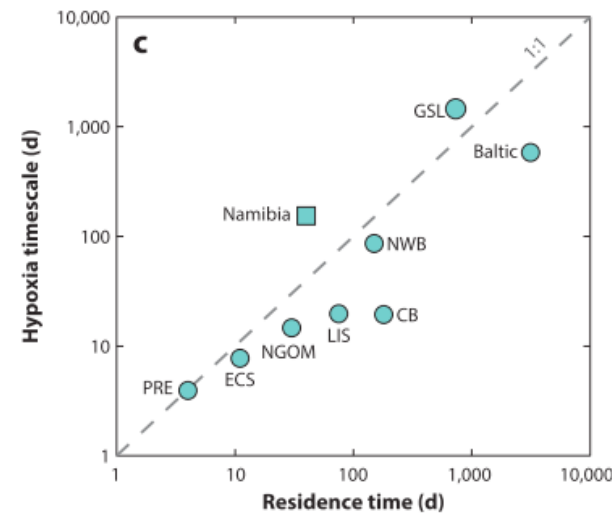
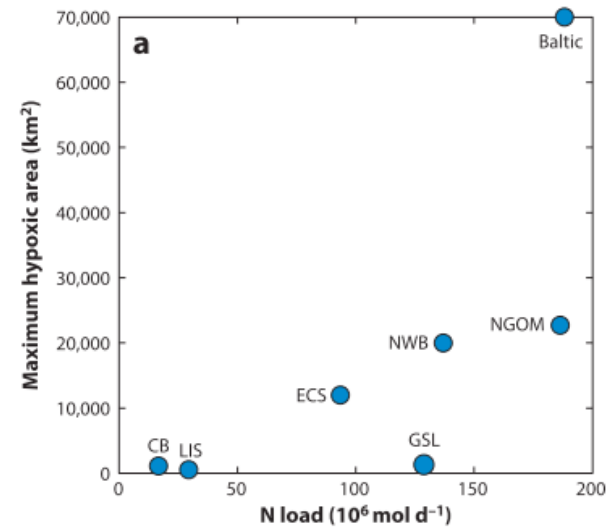
Seasonal stratification/hypoxia



Chesapeake Bay

Coastal hypoxia

- Higher nutrient load, longer residence time can lead to more hypoxia more hypoxia



Fennel and Testa, 2019

v2023

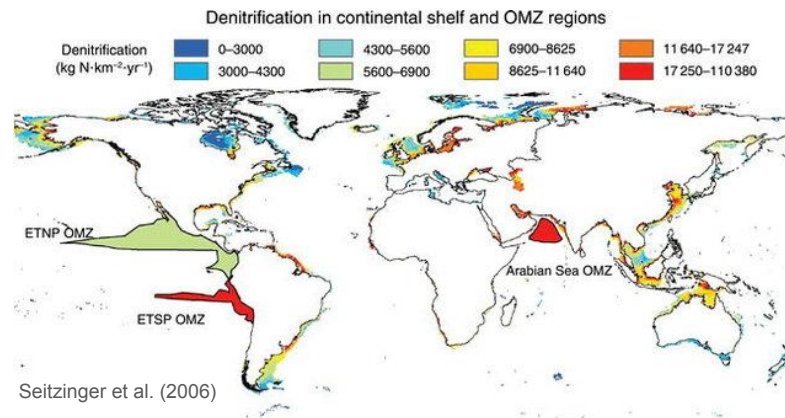
4.2 Nutrient cycling in estuaries

Denitrification

Table 1: Remineralization reactions (Mucci et al., 2000)

Reaction	ΔG (kJ/mol)
Aerobic respiration $(\text{CH}_2\text{O})_{106}(\text{NH}_3)_{16}\text{H}_3\text{PO}_4 + 138 \text{ O}_2 \longrightarrow \text{H}_3\text{PO}_4 + 16 \text{ HNO}_3 + 106 \text{ CO}_2 + 122 \text{ H}_2\text{O}$	-475
Denitrification $(\text{CH}_2\text{O})_{106}(\text{NH}_3)_{16}\text{H}_3\text{PO}_4 + 94.4 \text{ NO}_3^- \longrightarrow \text{HPO}_4^{2-} + 55.2 \text{ N}_2 + 106 \text{ HCO}_3^- + 71.2 \text{ H}_2\text{O} + 13.6 \text{ H}^+$	-448
Manganese reduction $(\text{CH}_2\text{O})_{106}(\text{NH}_3)_{16}\text{H}_3\text{PO}_4 + 236 \text{ MnO}_2 + 364 \text{ H}^+ \longrightarrow \text{HPO}_4^{2-} + 236 \text{ Mn}^{2+} + 8 \text{ N}_2 + 106 \text{ HCO}_3^- + 636 \text{ H}_2\text{O}$	-349
Iron reduction $(\text{CH}_2\text{O})_{106}(\text{NH}_3)_{16}\text{H}_3\text{PO}_4 + 424 \text{ Fe}_2\text{O}_3 + 756 \text{ H}^+ \longrightarrow \text{HPO}_4^{2-} + 424 \text{ Fe}^{2+} + 16 \text{ NH}_4^+ + 106 \text{ HCO}_3^- + 1060 \text{ H}_2\text{O}$	-114
Sulfate reduction $(\text{CH}_2\text{O})_{106}(\text{NH}_3)_{16}\text{H}_3\text{PO}_4 + 53 \text{ SO}_4^{2-} \longrightarrow \text{HPO}_4^{2-} + 53 \text{ HS}^- + 16 \text{ NH}_4^+ + 106 \text{ HCO}_3^- + 39 \text{ H}^+$	-77
Fermentation $(\text{CH}_2\text{O})_{106}(\text{NH}_3)_{16}\text{H}_3\text{PO}_4 \longrightarrow \text{H}_3\text{PO}_4 + 16 \text{ NH}_3 + 53 \text{ CO}_2 + 53 \text{ CH}_4$	-58

44% of global denitrification occurs in continental shelf sediments.



4.2 Nutrient cycling in estuaries

Cycling in sediments

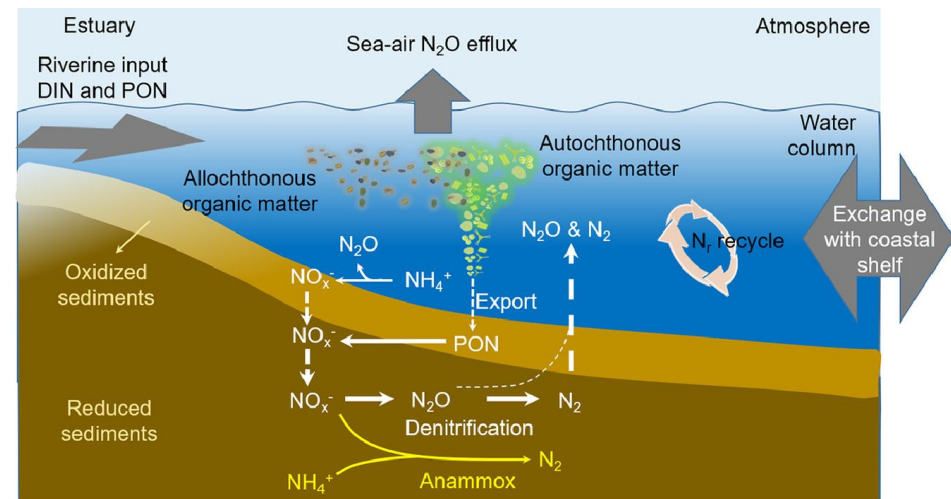
High production → High deposition rates

Shallow waters → More OM decomposition in sediments

→ Larger impact of nutrient release to the water column

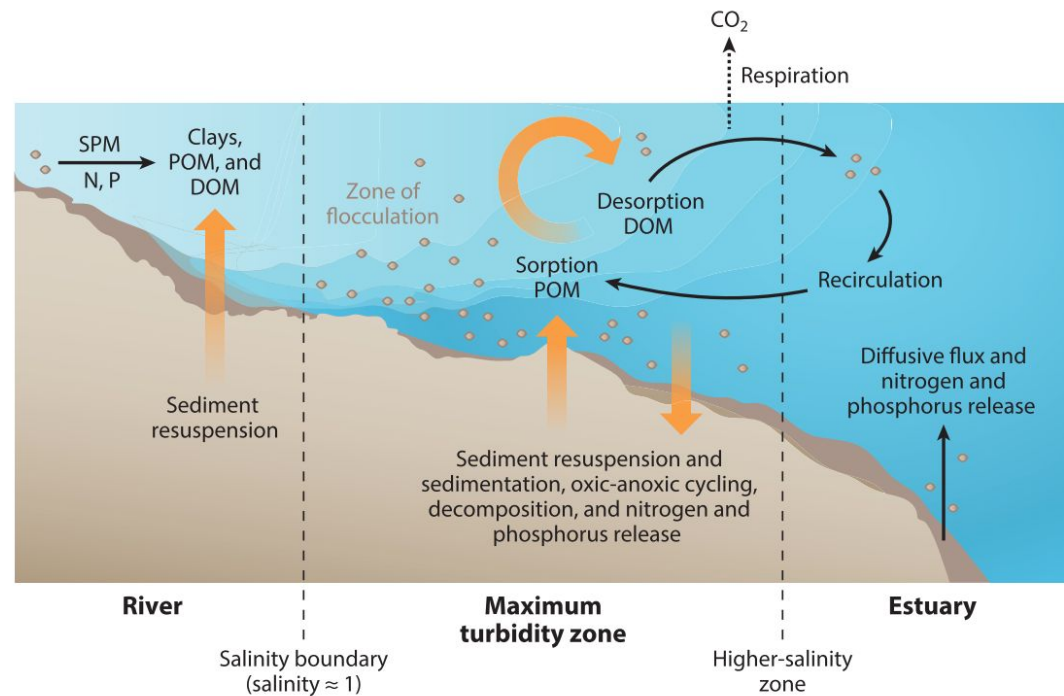
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Iron reduction $(CH_2O)_{106}(NH_3)_{16}H_3PO_4 + 424 Fe_2O_3 + 756 H^+ \rightarrow HPO_4^{2-} + 424 Fe^{2+} + 16 NH_4^+ + 106 HCO_3^- + 1060 H_2O$	-114
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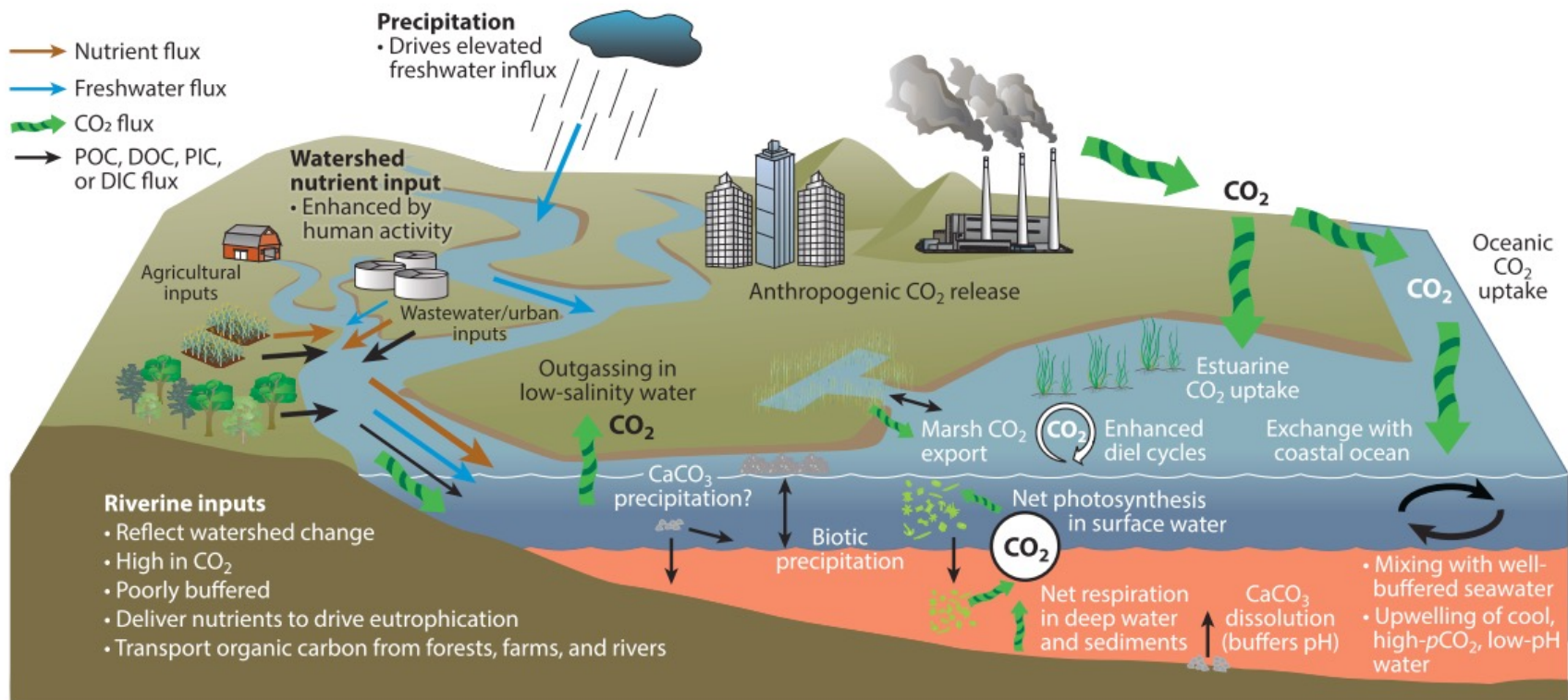
Tan et al. (2022)

4.2 Nutrient cycling in estuaries

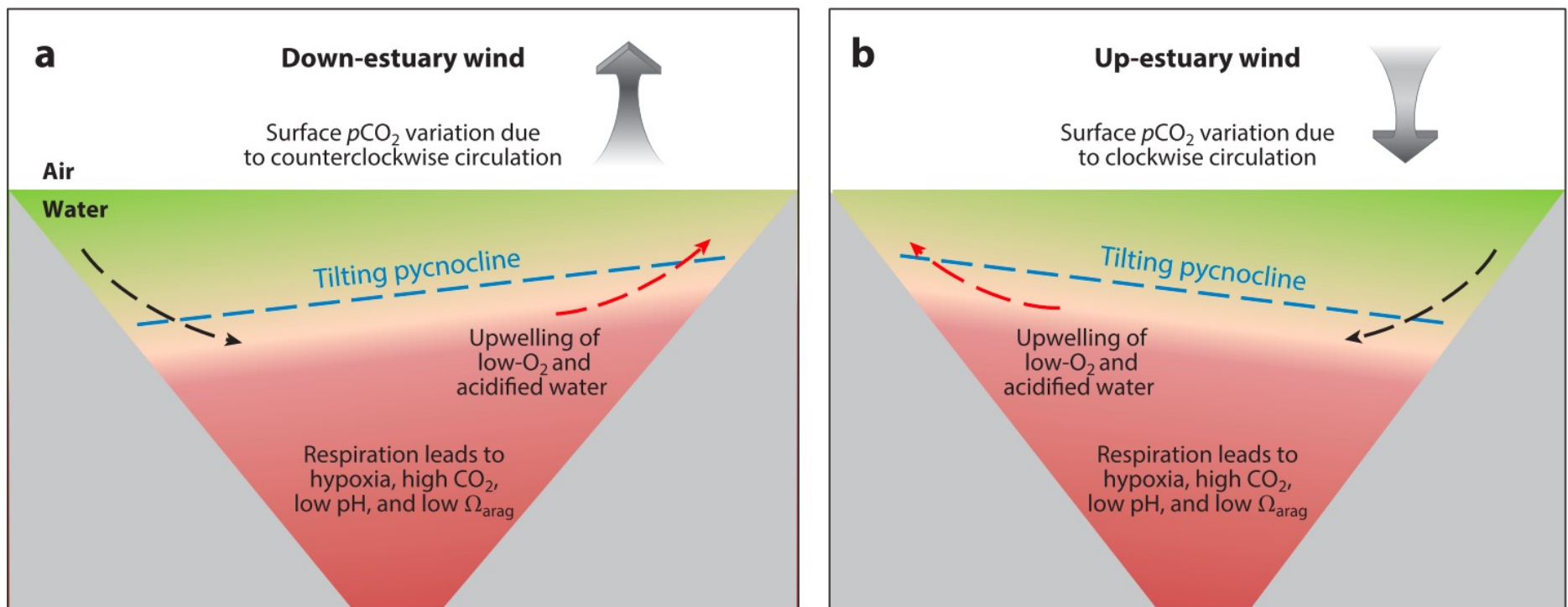


Canuel and Hardison, 2016

5. Acidification in coastal areas



5. Coastal hypoxia and acidification

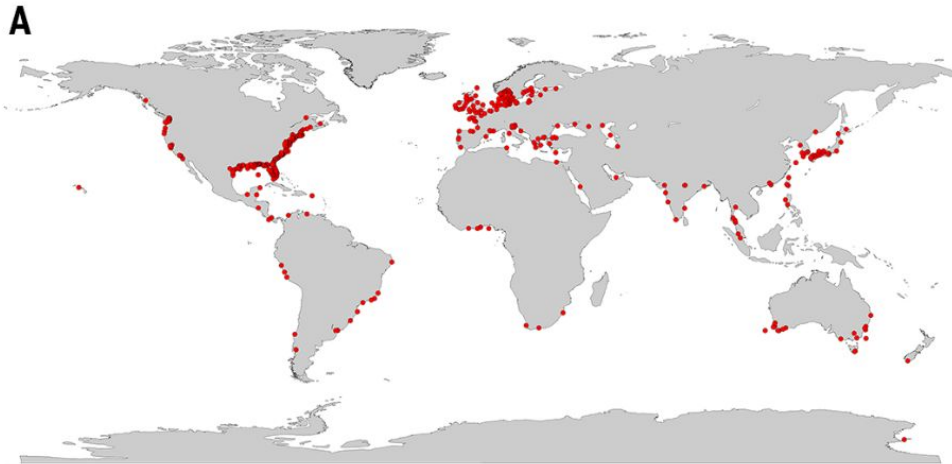


5. Current changes in coastal BGC

A. Enhanced anthropogenic export:

- Fertilizers ($\frac{3}{4}$ of the discharge)
- Sewage water ($\frac{1}{4}$ of the discharge)

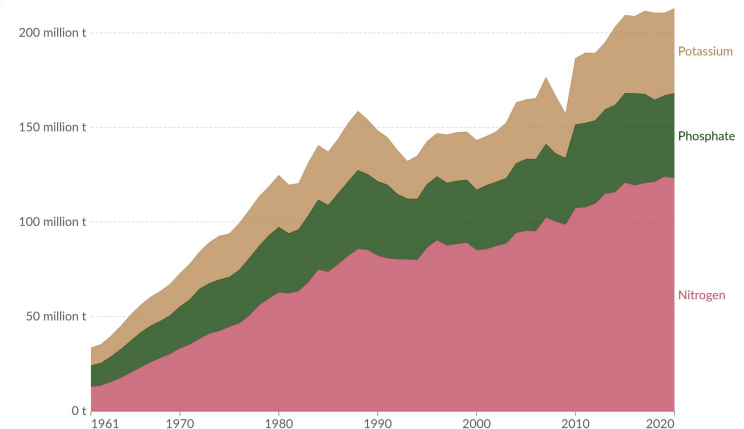
- Nitrogen discharge from rivers increased by 70% from 1860 to 1990 and continues to increase



Coastal waters where oxygen concentrations ≤ 61 mmol/kg have been reported (red). Breitburg 2018.

Fertilizer production by nutrient type, World, 1961 to 2020

Total fertilizer production by nutrient type (nitrogen, phosphate and potash), measured in tonnes of nutrient.



Data source: Food and Agriculture Organization of the United Nations

OurWorldinData.org/fertilizers | CC BY

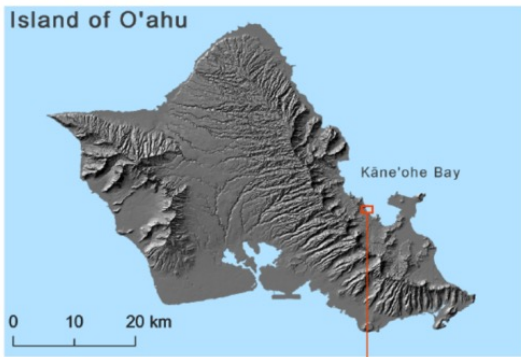
5. Current changes in coastal BGC

D. Physical changes

- a) Warming and enhanced stratification
- b) Sea level rise increases estuarine circulation
- c) Sea ice melt and enhanced mixing
- d) Changes in winds and Ekman pumping

Hawaiian fishponds – loko i'a

He'eia fishpond



nakilohonuaohoeia.org

- Sustainable sources of fish/food for thousands of years
- Significantly impacted by development, only a few (14) still in production (relative to hundreds pre-European contact)

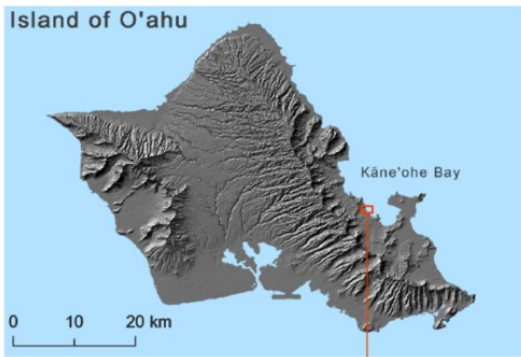


<https://paepaeohoeia.org/>



Hawaiian fishponds – loko i'a

He'eia fishpond



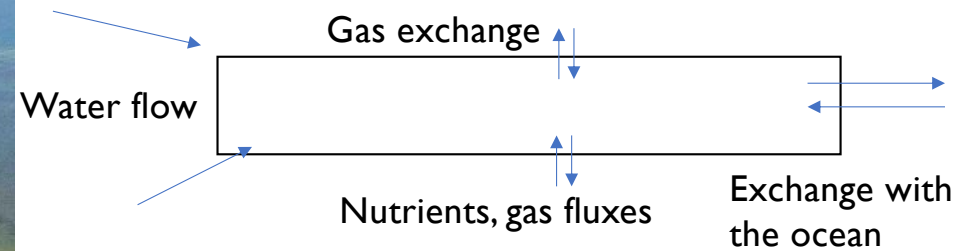
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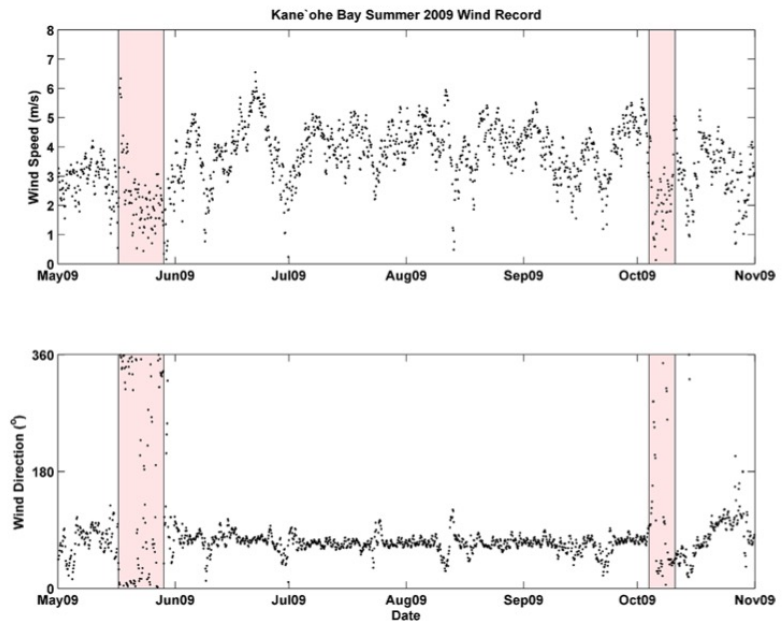
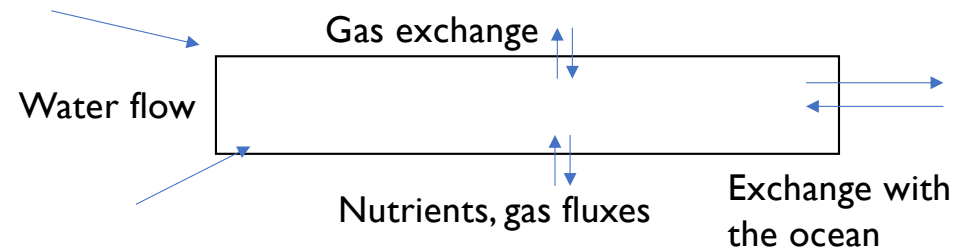
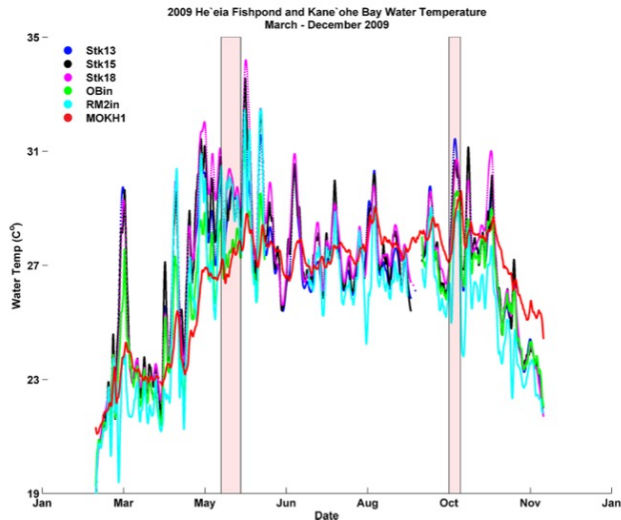
- System balances nutrient inputs, availability of oxygen, removal of wastes, accumulation of sediments
- Extreme conditions can develop quickly (small volume, shallow depth, large fluxes)



Hawaiian fishponds – loko i'a

He'eia fishpond

- Mortality events in 2009 – hypothesized link between change in wind direction / magnitude, high temperatures, hypoxic conditions (McCoy et al., 2017)
- No oxygen data to confirm



Hawaiian fishponds – loko i‘a

Modeling fish pond oxygen (and other biogeochemistry):
Widiasih, Garrido, Johnson, Kupihea 2021

$$\frac{dx}{dt} = P + A - S - R \quad (\text{Eq. 1})$$

x = oxygen concentration level in the water (in mg.L^{-1})

t = time (in hours),

P = Photosynthesis, A = Aeration,

S = Sedimentation, R =Respiration.

Matlab code available at: <https://osf.io/z2fh5/>

